

Exploring the Complementary Features of Audio and Text Notes for Video-based Learning in Mobile Settings

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ABSTRACT

In this paper, we compared audio and text note-taking behaviors for video-based learning in various mobile settings. We designed and implemented a note-taking tool and conducted a task-based study to examine how users took audio and text notes differently. Our results show that participants' audio notes were significantly longer than text notes; longer audio notes were taken to capture unfamiliar video content and participants' emotions. However, audio notes also raised several privacy concerns. Text notes allowed participants to revise for better accuracy and deeper reflection. Our findings of the complementary features of audio and text notes for video-based learning shed light on designing future note-taking tools that can be used to facilitate learning in varied mobile settings.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

KEYWORDS

Note-taking; Audio interaction; Mixed-method study

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1 INTRODUCTION

Efficient note-taking can promote learning, e.g., deepening people's comprehension of lecture content, promoting effective reading strategies, and improving the ability to synthesize information [21, 27, 35]. Scholars have developed note-taking functions for video-based systems, e.g., Collaborative Lecture Annotation System (CLAS) [32] and CourseMapper [7], to help students achieve learning objectives [15, 30]. It is suggested that note-taking should

be available across multiple platforms and in multiple formats for easy sharing and search [38] and students should be encouraged to take more structured and complete notes for effective learning [26]. However, note-taking for learning has been mostly researched in formal settings, such as in classrooms and formal online learning systems [14, 29]. Learning in informal settings differs from formal learning in many perspectives, e.g. process, location and setting, purpose, and content [8, 19]. Note-taking tools, such as Evernote, are used in various informal contexts, such as recreation, dining, or transportation [25, 29, 33, 40]. Research shows that note-taking in mobile contexts using wireless technology poses new challenges to efficient note-taking, such as being aware of learners' learning context, defined as the surrounding environment affecting learning services' delivery and execution [40].

Extensive works evaluated tools to support note-taking in different modalities (text, audio, images, etc.) and various contexts. For example, Schepman [33] found that note-taking software (e.g., Evernote) is used more frequently by mobile users than non-mobile users at different locations, suggesting that mobile note-taking tools are needed by students to supplement traditional platforms. However, few studied the factors that impact users' note-taking in different modalities (audio vs. text) when users are mobile [13, 33, 37].

To address this research gap, we designed and implemented a note-taking tool for people to take either audio or text notes when they watch videos. We ran studies to examine the research question: *How do people take text and audio notes in various mobility contexts when conducting video-based learning?* More specifically, we provided two versions of the note-taking tool (audio and text) and conducted a task-based study with 16 participants to gain deeper insights into if and how participants took audio and text notes differently in mobile and non-mobile contexts while watching videos and learning. The participants self-selected settings for them to take notes. We analyzed users' note-taking behavior using both quantitative and qualitative methods. The NASA-TLX [18] survey helped us better understand their mental workload for each task.

Our work makes the following contributions to the HCI community: 1) we designed and implemented a note-taking tool to examine the differences between audio and text note-taking; 2) we conducted a task-based study and provided empirical results on the complementary features (e.g., length of the notes, note content, and perceived cognitive load) of audio and text note-taking for video-based learning in multiple mobile settings; and 3) we provided several design implications for future note-taking tools for learning in informal settings where people may have different mobility.

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2 RELATED WORK

People conduct learning using diverse media content, e.g., videos and audio books [28], and under different mobility contexts. Ubiquitous learning tools allow users to easily adapt to different mobility contexts [40]. Most of the tools are developed to support note-taking in texts, e.g., Microsoft Word and Kindle, provide note-taking features through typing text [13]. Note-taking in other modalities has also been explored. For example, Han has shown that using images can help supplement text notes by providing additional information [17] and Towey has developed an experimental app that allows students to take notes using text, images, audio, and video [37]. Text has been used to summarize and tag audio recordings of business meetings [39]. Audio note-taking tools are becoming popular, given the convenience of recording and playback. Many tools are available in the market to support audio notes, e.g., Evernote. A major benefit of taking notes via voice lies in the fact that recording information by speaking requires fewer cognitive resources than writing, thus preserving the working memory capacity to perform additional tasks [16]. It is also shown that combining co-indexed handwritten notes and recorded audio in a digital notebook showed that such a system was more accessible than text notes alone and could enhance note-taking practices [39]. To create hands-free audio note-taking experiences, gaze tracking and machine learning techniques are combined to annotate PDF documents [23]. However, to the extent known, none compare audio and text notes that are taken by people when they watch videos under different mobility contexts. Among all media, video-based learning has been prevalent and effective [41]. Gaining a deeper understanding of the differences between the two note-taking modalities when people conduct video-based learning in informal settings can inform how to design more efficient note-taking tools.

Effectiveness of note-taking tools for learning in informal settings was evaluated on various measurements such as tool usability, e.g., a research paper designed a video learning note-taking tool to enhance personalized learning by outputting a timestamped hyperlink note document [12], and cognitive load, e.g., texSketch: Active Diagramming note-taking tools that combine pen-and-ink interactions with natural language processing to reduce the cost of producing diagrams while maintaining the cognitive effort necessary for comprehension [35]. Peer interaction is another important feature for note-taking for learning in informal settings, e.g. Livenotes: a shared whiteboard system and educational practice that uses wireless communication to impact cooperative learning in lectures [22].

3 METHOD

3.1 Interaction Design of the Note-taking Tool

We designed and implemented note-taking tools of two versions (audio and text) for video-based learning, as shown in Fig. 1. The text version allows people to type notes, and the audio version offers audio recording for note-taking.

The note-taking interface of the tool has three areas. At the top is the *Video View*, which displays the video content. The middle is the *Note Entry Panel*, which mainly consists of note-taking buttons that allow users to take notes with one click. For the Text version, the button triggers a note entry popup window for every note entry;

for the Audio version, the button triggers the audio recording until the user presses stop. Upon finishing entering/recording the note, a new entry of note pops up in the *Note View* at the bottom. On the right side, there is also a mute/unmute switch for users to promptly switch the video sound on and off if they would like to turn down the volume of the video. The intuitive interaction and large button allow users to take notes in a timely manner, such that audio note-taking can be easily accessed in different mobile contexts. The system also automatically mutes the video when they start recording audio notes. The users need to unmute the video once they finish note-taking to continue watching. We did not include a pause/start button since they would need to click two buttons to start taking notes and it is likely to increase note-taking effort.

The bottom *Note View* provides a list of the notes for participants to browse through. On the leftmost of each row of notes is a *Timestamp* button displaying the time in the video where the notes were taken. Clicking on the button plays the video from the moment the note was taken. In the Audio Version, clicking on the *Timestamp* button, the video jumps to the timestamp where the note was taken and starts playing the video. This feature builds coherency between original information and notes by linking the notes with the original video content. Besides, there is also an *Audio* button, which simply plays the audio note without interrupting the video from playing and aims to provide complete information on review. A *Label* button next to the play button enables the user to change the note to different colors to label the importance of the notes.

The *Note Content* label displays the note content. For the Text Version, the text is directly from what the user entered. For the Audio Version, the text is converted from the audio note by IBM Watson Speech Recognition Services. To deal with the latency of speech recognition, the tool displays a "Loading..." label in place of the note before the recognition is done. At the rightmost, a *Delete* button allows the user to delete the note at will. Participants could log into our tool any time using self-defined *Username* to review, edit and download their previous notes.

3.2 Task-based Study Design

In the study, participants used the note-taking tool we designed for taking notes while watching videos with 2x2 conditions (e.g., audio vs. text notes; mobile vs. non-mobile). Participants were given four videos to watch and were asked to take notes. Before starting note-taking tasks, researchers told participants to select contexts in which they have experienced the need to take notes. Participants were also told that they will be asked to share with researchers the most meaningful information from each video, while being informed they will not be assessed based on their memorization of the video content. The purpose for asking such a question is to make sure participants watched the videos. Exit interviews probe for details on how they took notes differently under the four tasks. By doing this, we aim to provide better comparisons of how note-taking behavior under different modalities (Audio vs. Text) would differ.

3.2.1 Note-Taking Tasks. The first session is the note-taking task session. We asked our participants to use two versions of our tool,

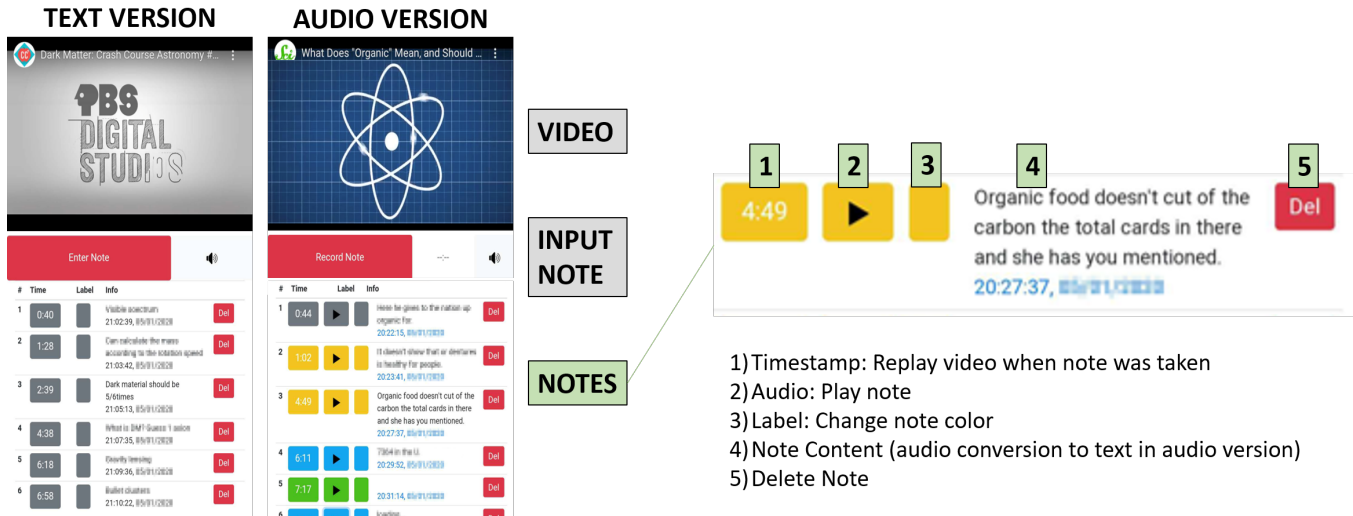


Figure 1: Sample user interface of our note-taking tool (Left: Text Version, Right: Audio Version). The major differences between the two versions are: 1) the Audio Version has a ‘play audio notes’ feature, and 2) the text next to the audio note button is automatically generated by speech-to-text recognition program on the fly.

the audio and the text version, to take notes in four self-defined learning scenarios. Among the four tasks, two were mobile contexts in which participants could move around to do other things that they felt comfortable with while watching the videos. The four tasks would be randomized to prevent the learning effect. We meant to compare audio/text notes by looking at note content/type, length, and location in the videos, thus we control the videos in the study by providing the same videos. To provide a better comparison between subjects, the research group pre-selected four STEM-related instructional videos from Youtube.com. Each video length was 10–11 minutes. The videos were from various topics and served as introductions on these topics for the wider public. Their names were “Dark Matter: Crash Course Astronomy - 41” [10], “Machine Learning & Artificial Intelligence: Crash Course Computer Science - 34” [11], “What Does “Organic” Mean, and Should You Buy Organic Foods?” [34], “The Power of Motivation: Crash Course Psychology - 17” [9]. They could take as many notes as they wanted, including none if they felt so inclined. After each video ended, we provided a link to fill in a NASA-TLX [18] survey to help us better understand their mental workload for each task. The colors are optional but not required for users to label notes after notes are taken. The notes taken by participants are logged by the system and shared with researchers. Statistical analysis was conducted on note quantity and note quality.

3.2.2 Exit Interview. During session one, researchers briefly organized participants’ data and proceeded to the post-study interview to understand their note-taking behaviors while using our tool. The researchers also asked the participants to share points they found helpful in each video while not looking at their notes. Our questions focused on participants’ note-taking strategies and actions in different tasks and reasons and experience for the four self-selected task scenarios, such as ‘for which task did you find note-taking to

be the most helpful and why?’, ‘do you remember any notes that were taken for it?’ We then ended the interview with open-ended questions on likes and dislikes about the current tool and how it could be further leveraged. The recordings were transcribed to text then analyzed using Affinity Diagrams [20] and LucidChart by two researchers in discussions. Our approach follows the constructing grounded theory approach [6].

3.2.3 Recruitment. We recruited 16 participants through emails and snowball sampling following the criteria : fluent in English and have an interest in audio note-taking. Among all 16 participants, seven of them were female and nine of them were male. Each study lasted between 90 minutes to 150 minutes. Each participant was compensated \$10-20 based on the time-length. Affected by the stay-at-home order, our in-person studies shifted online. It was conducted through video platforms such as Zoom or Skype. We recorded the video conferences with the consent of our participants.

4 FINDINGS

By analyzing the notes and connecting with exit interview results, we mainly focused on understanding the use of the two modalities for note-taking in different mobility contexts, including note quantity and cognitive workload. We found one common behavior: participants utilized the note-taking feature to annotate the structure and details. They also took notes to interpret video content. Audio recorded notes and text typed notes differed in terms of preferred context, quantity, and content.

Participants chose mobility contexts that aligned with their real-life scenarios. For example, participants chose to watch the videos when they did chores at home and physical exercises. Their self-selected mobility contexts are shown in Table 1. Given these contexts, we observed participants’ note-taking behavior and collected

Table 1: Phase II participants' demographic information and their self-selected mobility contexts.

ID	Gender	Age	Major	Self-Selected Mobility Contexts
P1	M	25-30	Engineering	Watch another entertaining video and organize household
P2	M	25-30	Kinesiology	Wash dishes, make coffee, microwave lunch
P3	F	25-30	Information Science	Clean bathroom and make juice
P4	F	20-25	Design & Engineering	Boil hot water, do laundry, eat snacks, and organize stuffs on the kitchen counter
P5	M	25-30	Engineering	Do Laundry, clean the kitchen counter
P6	F	25-30	Business	Organize household and wash dishes
P7	M	25-30	Information Science	Clean restroom and eating snacks
P8	M	25-30	Computer Science	Organize household and clean the house
P9	F	20-25	Engineering	Exercise and clean up the house
P10	M	25-30	Engineering	Organize household and wash dishes
P11	M	20-25	Math	Clean room
P12	F	25-30	Business	Make dessert and wash fruits
P13	M	20-25	Engineering	Eat fruit and make dessert
P14	F	25-30	Design	Cook hot tea and wash clothes
P15	M	25-30	Engineering	Wash clothes with hand and do laundry
P16	F	25-30	Crop Science	Clean the documents on the shelf and organize household

feedback on their experiences of using the tools. These findings suggested how people watched videos in different mobility contexts, doing varied daily activities in real life.

4.1 Audio Notes Expressed in Casual Language, Longer Than Text Notes

Research has shown that students who take more notes could learn better [24]. The number of words in student notes, which is positively related to better recall of ideas and details, has been used in educational research to measure student learning performance [3].

Overall, using the Audio version of the tool, participants on average took seven notes per video, and the length of the audio notes varied (mean=5.9, SD=4.8 words). Using the Text version of the tool, participants on average took nine notes per video, and the length of the audio notes varied (mean=4.6, SD=3.3 words). There were significant differences in length as represented by the word count data according to different modality conditions under a two-tailed repeated measures t-test, with the length of a note taken in the audio version being longer than the length of a note taken in the text version: $t(464) = -5.507, p < 0.0001$.

Text notes are often shorter than the audio note and contain keywords instead of full sentences. When the terminology was unfamiliar to the participants (e.g., "Axion" in the Astronomy topic video), both text and audio were used to take notes of the given term. However, audio notes usually contained the full statement (e.g., "Axion is one of the compositions of dark matter" (P16, Audio) vs. "Axion never be detected" (P14, Text). Another example in the Machine Learning Video is "Classification" (P3, Text) vs. "ok, so that's what a classifier means" (P5, Audio). We also observed that sometimes text notes contained multiple concepts or keywords that prolonged the text notes (e.g. "use manure in producing organic food and meat" (P11, Audio) vs. "Manure, less environmentally damaging, takes less energy to make chemical fertilizers" (P13, Text).

Compared to text notes, audio notes were longer. Participants' interviews show that they were not aware of such a difference when

comparing audio to text notes. But the participants did explain audio allowed them to take notes more naturally and flexibly. For example, a participant (P4) mentioned that the audio notes could contain more context than the text note. Some participants (e.g., P11, P13) also emphasized using such note-taking modality to repeat the video's content. They thought of audio notes as a way to focus more on the commentary contained in the video. For instance, when the video contained familiar content and the video explained a concept, they could use the audio function to repeat the key message. Additionally, participants felt that audio notes could help capture the emotional responses better than text. For example, P4 shared that she wanted to use the audio note-taking feature when watching dramas or accessing content that resonated more with emotions.

4.2 More Audio Notes for Unfamiliar Content

To better understand what notes participants took, we openly coded notes based on their relationship with the original video transcript. We defined three types of notes as (a) *Structural Information*: structural keywords and sentences that are directly mentioned in the video script; (b) *Detail Information*: non-structural keywords and sentences that are directly mentioned in the video script; (c) *Interpretation*: Importance, questions, comprehensions (not directly mentioned in original content), personal language. Due to audio to text auto-transfer accuracy, we had a type 'd' that included obvious errors and redundant notes, such as "testing." To focus on readable notes, we removed them in further quantitative analysis. For instance, notes like "an evolutionary perspective," "drive-reduction," "optima arousal," and "hierarchy of needs" are considered Structural Information; notes such as "social needs have to be balanced with our autonomy" are Detailed Information; and notes like "This is important," "T interact with M by gravity," "How do motivation help us," "Above is all build on statistic models" are Interpretations. During the open coding process, all audio notes were transcribed first using Watson Speech to Text, then manually transcribed by humans and compared with the machine transcribed texts. The accuracy of the text version for audio notes was increased like so.

All audio-converted-text and text notes were then coded separately by two researchers, and 76 out of 494 ($K = 0.154$) notes were coded as different types. These 76 notes were examined by the third researcher and coded as the outnumbered decisions, which left 19 ($K = 0.039$) notes uncategorized. Length for (a) *Structural Information*: mean = 4.8, sd = 3.8; (b) *Detail Information*: mean = 6.2, sd = 4.3; (c) *Interpretation*: mean = 5.5, sd = 4.2. The results suggest that the length of notes significantly differs from one type to another (X^2 -squared = 61.295, $p < 0.05$).

The types of these notes for each video were presented in Fig. 2. It showed that V2 (Dark Matters) received longer audio notes, 1.9 times that of its text notes; and V4 (Organic) received longer text notes, 2.2 times of its audio notes. According to the participants (P3, P4, P9, P13, P11, P16), V2 presented video content that they were not familiar with, thus they audio recorded many details to help capture new information. For example, P11 explained *"Compared to text, it's (audio) faster when recording full sentences to just take down whatever I hear. If text notes, I would need to further summarize the notes, which is difficult for an unfamiliar topic when you don't even know what is important."* On the other hand, the 'Organic' Video was related to the participants' everyday life and their familiarity with the topic allowed them to easily capture key-points of the video.

Some participants (e.g., P4, P9) mentioned that audio notes could help them express their emotions. For example, one participant (P9) commented that the audio note-taking mode helped them capture more context of the notes by recording the note content and their tone and other vocal features such as stutters. P4 shared that she used these signals to remind her of exciting points in videos. Meanwhile, privacy concerns of taking audio notes were also shared by seven participants. It was not about being overheard while taking down audio notes since people would not even try to take audio notes when people they know are around. Instead, they were concerned about reviewing or sharing the audio notes with others. *"Audio is identifiable data. Someone is more likely to misuse my audio data than my text notes (P13)."* *"Audio notes create a character, it shows who you are and what you like (P1)."* *"It is embarrassing to show others your (audio notes) (P10)."*

However, when participants made text notes, they tended to put additional effort into summarizing their thoughts in shorter forms and made sure the spelling and grammar were all correct. The text notes had more *"in-depth"* thoughts. As some participants mentioned, it took more time to polish the text notes before finishing entering, e.g., P14 spent more time to make sure the note *"looks right"* but didn't do so when recording audio notes.

4.3 Cognitive Load of Note-Taking Modality Perceived Differently by Individuals

When using MANCOVA to model all five NASA scales, Mental Demand, Temporal Demand, Effort, and Frustration, the results suggest that the only significant relevant independent variable is Participant ID ($F(80, 180) = 2.249$, $p < 0.001$, Pillai = 2.333). In other words, the cognitive load scores varied by participants. The audio version has higher means than the text version on Frustration; however, the difference was not statistically significant. Overall, though there were many factors, tool modality and context mobility

do not significantly impact people's perceived load of note-taking, and it varied by individuals.

Across these mobility scenarios, all of our participants commented on the convenience of audio note-taking as it allowed them to enter notes while working on the other task at the same time. Participants mentioned that the audio note-taking mode allowed them to speak their mind directly. Typing was less feasible in these contexts because their hands were occupied with other tasks. One participant remarked: *"It's really difficult for me to use text while I'm doing something. If I'm doing something, I would prefer to use audio (P5)."*

Though audio note-taking seemed to offer better accessibility, participants also shared some challenges of taking audio notes. The environment for taking audio notes needs to be quiet. They had to make sure the system could correctly record the notes and need to be able to hear the sound of the video to learn. Participants also preferred to audio record in a quieter environment, in which they could be more focused. The self-selected activities sometimes distracted the participants, and they had to stop watching the videos. In such cases, they look for solutions that better support the switches between activities or to focus on one at a time. For instance, one participant (P9) mentioned that as she could not focus on two things at the same time, she had to put down the other tasks that she was doing and focus on finishing the video and audio notes.

Participants also shared that the audio notes had more accuracy issues due to the speech-to-text recognition, which caused much inconvenience, even though they praised the audio note-taking feature for entering notes more efficiently. Some instantly reviewed these recognized texts after they were generated. If the text was not as accurate as they expected, some took additional effort to delete and redo the notes or put up with obscure words. This additional effort compromised the convenience of audio note-taking. One participant (P16) particularly mentioned that the audio notes recorded and converted to text inaccurately making it unusable and drew away attention from the content. P16 could not remember the reason why she took such a note upon reviewing: *"Sometimes what I said would not be recognized to texts correctly. When I tried to review the note, I can only use part of the sentence that's recognized to recall what my point was. If the whole sentence is unreadable, I will not remember what the point was."*

5 DISCUSSION AND CONCLUSION

Extensive literature has been dedicated to the design of mobile note-taking tools [33, 36]. Notes can be taken in different modalities, audio, text, images, and videos, etc. However, none compared audio and text note-taking using videos in different mobile contexts. Our work contributes to new findings about the complementary roles these two types of notes make for video-based learning and make several design implications for more efficient note-taking tools.

First, we observed that in most self-selected mobile contexts, participants' hands were occupied with other tasks, such as washing, exercising, and cleaning. Further mobile note-taking tools could detect contextual factors to set a default note-taking modality, e.g., offering audio note-taking features while detecting the users are riding a bike or taking a bus [5], and integrating human typed

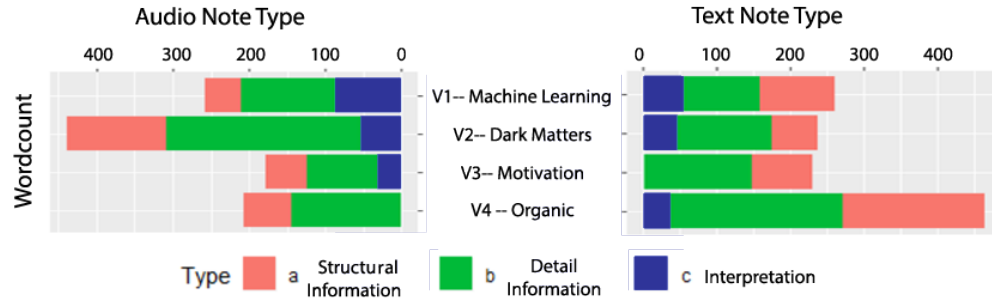


Figure 2: Total audio and text notes for the four videos. Note types vary by color.

notes with descriptive video captions to leverage the note-taking experience [1]. By reducing the need to focus on note-taking mechanics, note-taking tools could better support students with specific learning disabilities [2]. However, inaccurate speech-to-text generation could result in users' frustration and additional efforts for correcting the notes. Participants' feedback about a relatively quiet environment is needed for taking audio notes further and can be used to improve recommendations, e.g., sound sensors could be used to evaluate the surrounding noise level and text notes may be recommended when it is too noisy to make audio recordings.

Second, our findings showed that audio notes could be used to capture users' emotions and improvised thoughts naturally; on the contrary, participants tended to revise text notes to be concise and accurately represent participants' ideas. Participants had more privacy concerns about sharing audio notes with others, even after the audio notes were converted to text. Knowledge sharing is important for improving learning in mobile and ubiquitous learning, e.g., [19]. In future designs of audio note-taking tools, speech to text algorithms could be applied to remove parts of the notes (e.g., having personal language or emotion-related content) that may be more privacy sensitive, such that people may be more willing to share notes.

Last, technologies provide learners with greater independence, individualized instruction, and greater control over their learning experiences [4]. Our results also showed the perceived cognitive load varied by individual participant, instead of the note-taking modality or if the notes were taken in a mobile/non-mobile context. Considering that text is more accessible for deaf or hard-of-hearing users and audio is more approachable for people who have vision challenges, the complementary characters of the two note-taking modalities can be leveraged to promote accessibility. For example, future learning systems are suggested to support both modalities for taking notes, and notes taken in one modality may need to be automatically converted to the other format for easy access across varied mobile settings.

In future studies, a deeper analysis than the note length and note content can be done to further understand the information differences between audio and text notes, such as linguistic formatting in note-taking [31]. Our participation pool is exclusively students; further studies may look into other user populations and among a bigger sampling size.

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